Nonlinear Programming: Midterm Review

1 Running example: wildfire treatment

Review points: formulation

- **LP versus NLP:** an NLP has a nonlinear objective or at least one nonlinear constraint. Marginal benefits or resource requirements can vary with decisions.
- **Model components:** identify decision variables and units, the objective, resource limits, and sign restrictions.
- **Feasibility versus optimality:** constraints determine which choices are possible; the objective ranks those choices.

Apply: define the treatment choices.

A land-management agency chooses prescribed burning, b, and mechanical thinning, m, on separate eligible areas. Each unit is a **10-acre treatment block**; fractional blocks are allowed. The agency treats the highest-priority sites first, so additional blocks generate diminishing marginal net benefits. For this hypothetical exercise, net benefits are

$$N(b, m) = 10b - b^2 + 10m - m^2.$$

Net benefits are measured in **units of \$10,000** and represent expected avoided wildfire losses minus treatment costs, evaluated over the same planning period. The two areas' benefits are additive; this example excludes interactions such as fire spreading between treatment areas. The quadratic functions are stylized approximations, not estimates of treatment effectiveness.

Treatment	Decision (10-acre blocks)	Cost per block	Net-benefit contribution
Prescribed burning	b	\$10,000	$10b - b^2$
Mechanical thinning	m	\$20,000	$10m - m^2$

With budget B, also in \$10,000 units, spending must satisfy $b + 2m \leq B$. Costs reduce net benefits **and** use available funding. These are different roles, so including costs in the objective and a spending constraint does not double-count them. The agency maximizes net benefits, not acres treated.

2 Unconstrained optimization

Review points: FOCs, SOCs, and global optimality

- **Interior FOC:** for a differentiable objective, $f'(x^*) = 0$ in one dimension and $\nabla f(x^*) = 0$ in several dimensions. Every partial derivative must equal 0.
- **One-variable SOC:** at a stationary point, $f'' < 0$ gives a strict local maximum, $f'' > 0$ a strict local minimum, and $f'' = 0$ is inconclusive.
- **Multivariate SOC:** use the full Hessian, including cross derivatives. A negative-definite Hessian gives a strict local maximum; positive definite gives a strict local minimum; indefinite gives a saddle point.
- **Two-variable test:** for a symmetric Hessian, $f_{11} < 0$ and $\det(H) = f_{11}f_{22} - f_{12}^2 > 0$ establish negative definiteness. In higher dimensions its leading principal minors alternate signs, starting negative.
- **Local versus global:** a local maximum dominates nearby feasible choices; a global maximum dominates all feasible choices. Concavity on a convex domain makes an interior stationary point a global maximizer; strict concavity ensures uniqueness. Stationarity alone proves neither.

First ignore the budget and solve over real-valued choices. The resulting positive choices will also satisfy the physical nonnegativity restrictions.

2.1 Solve without a budget

For burning alone,

$$\max_b n(b) = 10b - b^2, \qquad n'(b) = 10 - 2b = 0 \quad \Rightarrow \quad b^* = 5.$$

Since $n''(b) = -2 < 0$ everywhere, this is the unique global maximum. Below five blocks, another block has positive marginal net benefits; above five, it has negative marginal net benefits.

With both treatments,

$$\nabla N = \begin{bmatrix} 10 - 2b \\ 10 - 2m \end{bmatrix}, \qquad H = \nabla^2 N = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}.$$

The FOCs give $(b^u, m^u) = (5, 5)$. Net benefits are 50 (\$500,000), and spending is $5 + 2(5) = 15$ (\$150,000). Each treatment covers 50 acres.

The leading principal minors are $-2 < 0$ and $4 > 0$, so H is negative definite. Because this holds everywhere, N is strictly concave, establishing a unique global maximum.

3 Numerical methods: finding a zero marginal net benefit

Review points: optimization as rootfinding

- Solve the FOC by finding a root of $g = f'$, or of ∇f with several decisions. A root of the objective itself is not the optimization target.
- A numerical root identifies a stationary candidate; check curvature and feasibility before calling it an optimum.

Here $g(b) = n'(b) = 10 - 2b$. We know the answer analytically and can use it to check each algorithm.

3.1 Bisection: maintain a bracket

- **Requirements:** a continuous root function and endpoints with opposite signs. A root without a sign change can be missed; multiple roots need not be distinguished by one bracket.
- **Update:** evaluate the midpoint and retain the half with a sign change. The bracket width halves each iteration (linear convergence).
- **Stopping:** width at most 2ϵ places the midpoint within ϵ of a bracketed root. Also inspect the function residual.

Start with $[a, c] = [0, 8]$. Continuity and $g(0) = 10 > 0 > g(8) = -6$ bracket a root. At each step evaluate the midpoint, then retain the half with a sign change.

Step	Bracket	Midpoint	g(midpoint)	Next action
1	$[0, 8]$	4	2	Keep $[4, 8]$.
2	$[4, 8]$	6	-2	Keep $[4, 6]$.
3	$[4, 6]$	5	0	Stop: exact root.

3.2 Gradient ascent is a fixed-point iteration

- **Fixed point:** rewrite the equation as $x = T(x)$ and iterate $x^{k+1} = T(x^k)$. For a continuously differentiable map, $|T'(x^*)| < 1$ gives local convergence from sufficiently nearby starting points.
- **Gradient direction:** use $x^{k+1} = x^k + \alpha_k \nabla f(x^k)$ for maximization and a minus sign for minimization. The gradient gives steepest local increase under Euclidean distance; the step size controls how far to move.
- **Step size matters:** too small can make progress slow; too large can cause oscillation or divergence. Different fixed-point rearrangements can converge differently even when they have the same solution.

Use marginal benefits to choose an adjustment:

$$b^{k+1} = b^k + \alpha g(b^k) = T(b^k), \qquad T(b) = (1 - 2\alpha)b + 10\alpha.$$

With $\alpha = 1/4$ and $b^0 = 0$, the sequence is

$$0 \longrightarrow 2.5 \longrightarrow 3.75 \longrightarrow 4.375 \longrightarrow \cdots \longrightarrow 5.$$

Writing $e^k = b^k - 5$ gives $e^{k+1} = (1 - 2\alpha)e^k$. Thus this iteration converges from any starting point when $|1 - 2\alpha| < 1$, or $0 < \alpha < 1$. At $\alpha = 1$, errors oscillate without shrinking; larger positive steps diverge unless started exactly at the solution.

3.3 Newton: use curvature to scale the adjustment

- **Rootfinding update:** $x^{k+1} = x^k - g(x^k)/g'(x^k)$. For optimization, substitute $g = f'$ to obtain $x^{k+1} = x^k - f'(x^k)/f''(x^k)$.
- **Convergence:** under suitable smoothness, a simple root, and a sufficiently close starting point, Newton converges quadratically. It is a local guarantee.
- **Failure modes:** a small or zero denominator, a poor starting value, or an inappropriate step can prevent convergence or lead to the wrong stationary point.

Apply Newton to $g = 0$:

$$b^{k+1} = b^k - \frac{g(b^k)}{g'(b^k)} = b^k - \frac{10 - 2b^k}{-2} = 5.$$

Newton reaches the answer in one step because g is linear.

4 KKT: evaluate different budget levels

Review points: inequalities and active constraints

- **Lagrangian:** for maximization with $g(x) \leq B$, use $\mathcal{L} = f(x) + \lambda[B - g(x)]$ with $\lambda \geq 0$. An equality multiplier, by contrast, is unrestricted in sign.
- **Complementarity logic:** slack constraint $\Rightarrow$ zero multiplier; positive multiplier $\Rightarrow$ binding constraint. A binding constraint can have a zero multiplier.
- **Case analysis:** hypothesize the binding constraints, solve the resulting equations, and reject candidates violating any primal or dual condition.

For any budget $B > 0$, write the complete problem:

$$\max_{b,m} \; N(b,m) \quad \text{s.t.} \quad b + 2m \leq B, \quad b, m \geq 0,$$

Lagrangian:

$$\mathcal{L} = 10b - b^2 + 10m - m^2 + \lambda(B - b - 2m) + \eta_b b + \eta_m m$$

We will apply the same conditions at $B = 20$, $B = 5$, and $B = 2$.

4.1 Full KKT conditions in inequality form

- **Stationarity with nonnegative choices:** each choice derivative of $\mathcal{L}$ is nonpositive, and equals zero whenever that choice is positive.
- **Primal feasibility:** both choices and the budget slack are nonnegative.
- **Dual feasibility:** the budget multiplier is nonnegative.
- **Complementary slackness:** each choice times its Lagrangian derivative is zero; the budget multiplier times unused funding is zero.

Written out in full, these conditions are

$$\begin{aligned}
\mathcal{L}_b = 10 - 2b - \lambda \leq 0, \quad & b \geq 0, \quad & b(10 - 2b - \lambda) = 0, \\
\mathcal{L}_m = 10 - 2m - 2\lambda \leq 0, \quad & m \geq 0, \quad & m(10 - 2m - 2\lambda) = 0, \\
\mathcal{L}_\lambda = B - b - 2m \geq 0, \quad & \lambda \geq 0, \quad & \lambda(B - b - 2m) = 0.
\end{aligned}$$

4.2 Loose budget: $B = 20$ ($200,000)

The unrestricted choice $(5, 5)$ spends $15 < 20$.

4.3 Moderate budget: $B = 5$ ($50,000), both treatments positive

Assume $b, m > 0$. The conditions $b\mathcal{L}_b = 0$ and $m\mathcal{L}_m = 0$ force $\mathcal{L}_b = \mathcal{L}_m = 0$, reducing the KKT system to

$$10 - 2b - \lambda = 0, \qquad 10 - 2m - 2\lambda = 0, \qquad b + 2m = 5.$$

The stationarity equations give $b = 5 - \lambda/2$ and $m = 5 - \lambda$. Insert these into the binding budget:

$$\left(5 - \frac{\lambda}{2}\right) + 2(5 - \lambda) = 5 \quad \Rightarrow \quad \lambda^* = 4, \qquad (b^*, m^*) = (3, 1).$$

Check all four conditions: both marginal inequalities hold at equality, $10 - 6 - 4 = 0$ and $10 - 2 - 8 = 0$; spending is $3 + 2(1) = 5$ with positive choices; $\lambda = 4 \geq 0$; and all three complementary products are zero. Net benefits are $N(3, 1) = 30$ ($300,000).

4.4 Hessian check for the positive-treatment solution

Review points: second-order conditions

- **Constrained SOC:** at a regular equality-constrained stationary point, negative curvature in every nonzero tangent direction implies a strict local maximum. The bordered Hessian incorporates the active constraint.

For the decision vector $x = (b, m)^\top$,

$$H = \nabla^2_{xx}\mathcal{L} = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}.$$

Its leading principal minors are $-2 < 0$ and $(-2)^2 - 0 = 4 > 0$, so it is negative definite.

$$H_B = \begin{bmatrix} 0 & -1 & -2 \\ -1 & -2 & 0 \\ -2 & 0 & -2 \end{bmatrix}, \qquad \det(H_B) = 10.$$

The principal minors of H_B alternate signs:

$$\det[0] = 0, \quad \det\begin{bmatrix} 0 & -1 \\ -1 & -2 \end{bmatrix} = -1 < 0, \quad \det(H_B) = 10 > 0.$$

Starting with the first nonzero minor, the signs alternate, so H_B is negative definite.

4.5 Tighten the budget to $B = 2$ ($20,000)

Case 1: binding budget, both treatments positive. Complementary slackness requires $\mathcal{L}_b = \mathcal{L}_m = 0$. As before,

$$b = 5 - \lambda/2, \qquad m = 5 - \lambda, \qquad 15 - \tfrac{5}{2}\lambda = 2.$$

This gives $\lambda = 26/5$, $b = 12/5$, and $m = -1/5$. The negative thinning choice violates primal feasibility. Reject the assumption of two positive treatments.

Case 2: burning positive, thinning zero. Set $m = 0$. The binding budget gives $b = 2$. Because $b > 0$, complementary slackness requires $\mathcal{L}_b = 0$:

$$10 - 2(2) - \lambda = 0 \quad \Rightarrow \quad \lambda = 6,$$

Because $m = 0$, the thinning condition remains an inequality:

$$\mathcal{L}_m = 10 - 2(0) - 2(6) = -2 \leq 0, \qquad m\mathcal{L}_m = 0(-2) = 0.$$

Therefore

$$\boxed{(b^*, m^*) = (2, 0), \qquad \lambda^* = 6.}$$

In the equivalent explicit-multiplier form, $\eta_b = 0$ and $\eta_m = 2$.

4.6 Verify the accepted solution

Condition	Check at $(b, m, \lambda) = (2, 0, 6)$
Marginal inequalities	$\mathcal{L}_b = 10 - 4 - 6 = 0 \leq 0$; $\mathcal{L}_m = 10 - 0 - 12 = -2 \leq 0$.
Primal feasibility	$2 + 2(0) = 2$; both treatments nonnegative.
Dual feasibility	$\lambda = 6 \geq 0$.
Complementary slackness	$2(0) = 0$; $0(-2) = 0$; $6(2 - 2) = 0$.

Net benefits are $N(2, 0) = 16$ ($160,000). The agency burns 20 acres and funds no mechanical thinning. This conclusion follows from these hypothetical returns, costs, and budget; it is not a general ranking of wildfire treatments.

5 See the three allocations

- **Contour interpretation:** move toward higher objective contours while remaining feasible. The gradient is perpendicular to an objective contour.
- **Location of an optimum:** an NLP can attain its optimum in the interior, along an edge, or at a corner. Checking only vertices can miss the answer.
- **Binding versus valuable:** a boundary limits feasible choices; its multiplier measures the marginal value of relaxing it. These are distinct concepts.

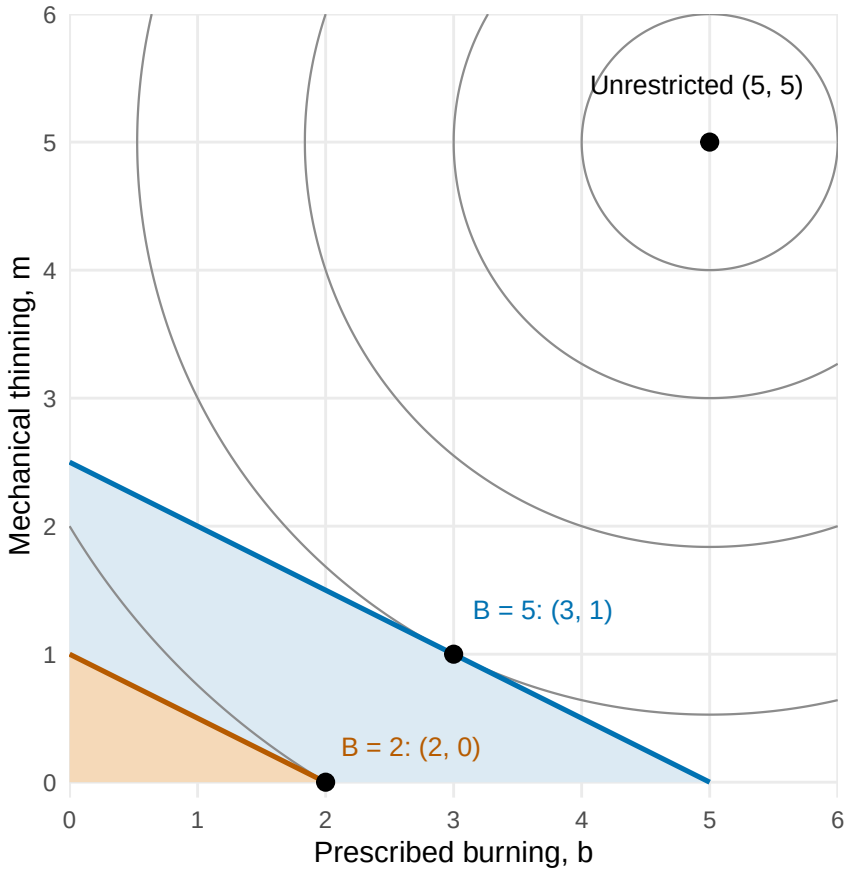


Figure 1: Net-benefit contours and two budget sets. The moderate-budget optimum is on an edge with both treatments positive. The tight-budget optimum is a corner where thinning is zero. Axes are 10-acre blocks.

Budget B	Burning b^*	Thinning m^*	Net benefits N^*	Budget multiplier
Unrestricted	5	5	50	–
20 (slack)	5	5	50	0
5 (binding)	3	1	30	4
2 (binding, corner)	2	0	16	6